SUPERVISOR REVIEW OF MASTER'S THESIS

Master's student: Assanova Sandugash

Educational program: Applied Data Analytics

Thesis title: Development of Machine Learning Models to Improve the Predictive Ability and Accuracy of Forecasting Annual Medical Care Volume

1. Relevance of the Research Topic

The topic of the master's thesis is relevant because annual healthcare expenditure forecasting is a core planning function for health ministries, yet machine learning approaches have not been systematically evaluated for this purpose in the Kazakhstan context. The country's health system has undergone major structural reform and experienced a significant fiscal shock during the COVID-19 pandemic, making it a particularly demanding forecasting environment. The research addresses a documented gap in the literature and produces results that are directly applicable to the Ministry of Health's budget planning processes.

2. General Characteristics of the Thesis

The master's thesis consists of an introduction, five chapters, a conclusion, a list of references, and appendices containing fully reproducible Python code. The thesis is well-structured, and the appendix code is a notable strength: it allows all reported results to be reproduced from the original data files without manual intervention.

3. Scientific Novelty

The scientific novelty of the work lies in the first comparative benchmark of classical ARIMA and ensemble machine learning methods for national-level healthcare expenditure forecasting in Kazakhstan; in the two-track evaluation design that separately assesses model performance under adequate sample conditions and under extreme data scarcity; in the documentation of the multicollinearity structure of Kazakhstan's health system indicators and its consequences for model stability and feature attribution; and in the demonstration that the regional administrative panel can serve as a practical ML training corpus when the national time series is too short for direct ensemble learning.

4. Brief Description of the Thesis

The introductory chapter defines the research problem, objectives, hypotheses, and scope. The literature review covers classical time-series methods, ensemble machine learning for healthcare forecasting, temporal validation methodology, and the Kazakhstan planning context. The methodology chapter describes the six source datasets, preprocessing steps, exploratory analysis, feature engineering for both analytical levels, model specifications, and the full evaluation protocol, including a walk-forward procedure introduced to address the evaluation asymmetry between ARIMA and ML methods. The results chapter presents metrics across both tracks, an ablation study, feature importance analysis with permutation testing, multicollinearity diagnostics, inflation sensitivity analysis, and three-year forecasts with bootstrap uncertainty bands. The discussion and conclusion chapters interpret the findings against the stated hypotheses and translate them into concrete recommendations for health planning.

5. Student Performance Evaluation

During the preparation of the thesis, the student demonstrated independence, methodological care, and the ability to work through a genuinely difficult forecasting problem involving short time series, high multicollinearity, and two distinct analytical levels. Of particular note is the student's initiative in introducing the expanding-window walk-forward evaluation for

ARIMA – an addition that was not part of the original research design but that substantially improves the integrity of the model comparison. The student responded constructively to feedback throughout the supervision period, revised the analysis where the initial results required more careful interpretation, and was consistently transparent about the limitations of the findings rather than overstating them. The work demonstrates readiness for professional or research practice in applied data analytics.

6. Conclusion

The master's thesis is completed in full, meets all the requirements of the Applied Data Analytics programme, and can be admitted to the defense. The work deserves the grade «A».

Supervisor: Beibit Abdikenov, PhD, Director of Science and Innovation Center «Artificial Intelligence»

Signature: ____________ Date: ____________